

TOPIC:



Gateway® Technology

The Functional Genomics Operating System

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GATEWAY CLONING

- ◉ The **Gateway cloning System**, invented and commercialized by Invitrogen life technologies since the late 1990s.
- ◉ It is a molecular biology method that enables researchers to efficiently transfer DNA-fragments between plasmids using an appropriate set of recombination sequences, the "Gateway att" sites, and two proprietary enzyme mixes, called "LR Clonase", and "BP Clonase"

GATEWAY CLONING

- ◉ Gateway cloning is universal cloning method, based upon site-specific recombination properties of bacteriophage lambda which facilitates the integration of lambda into *E. coli* chromosome and switches between lytic and lysogenic pathways.
- ◉ Gateway technology provides a rapid, efficient, accurate way of moving DNA sequences into multiple vectors for functional analysis and protein expression.

GATEWAY CLONING

- ◉ Gateway technology is totally based upon recombination reaction.
- ◉ RECOMBINATION COMPONENT:
Lambda based recombination involves two components:
 1. The DNA recombination sequences (att sites) four att sites.
 2. Enzyme that mediate recombination reaction(i.e. clonase enzyme mix)

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BP Reaction

From
lambda

attP



attB

From Ecoli
or
produces
in PCR-
Product

By
combination
of *attB* and
attP

attL



attR



By
combination
of *attB* and
attP

LR Reaction

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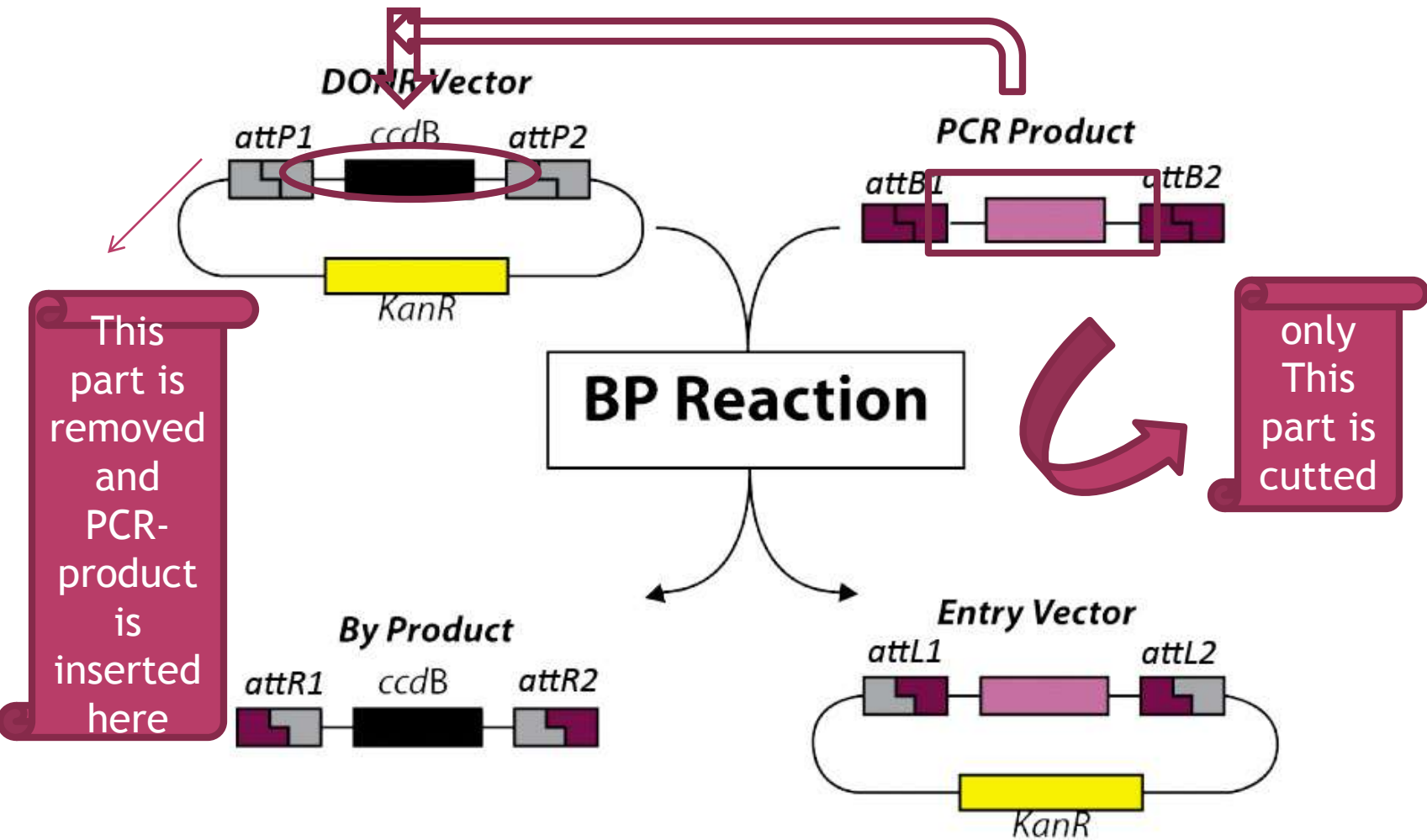
- Recombination reaction:

There are two types of recombination reaction.

- 1. BP reaction:



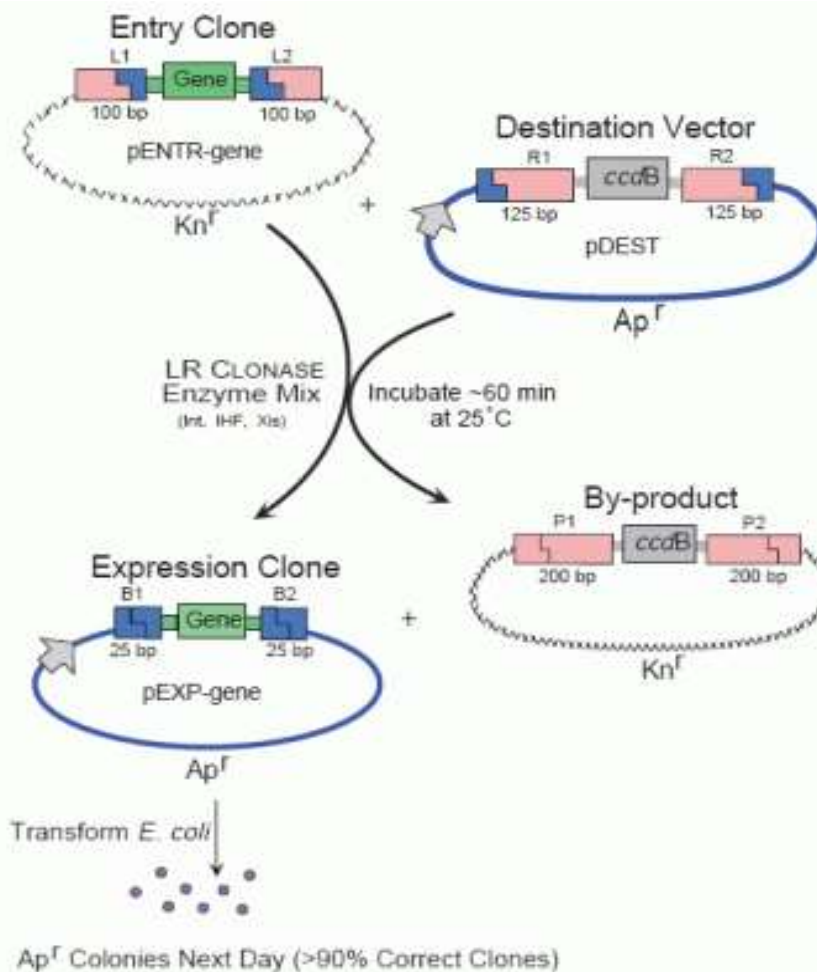
GATEWAY CLONING



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2. LR reaction

attL * attR $\xrightarrow{\text{LR clonase mix}}$ attB * attP



Colonies containing expression clone

GATEWAY CLONING

- ◉ Features of recombination reaction.
- ◉ **ccdB gene :**
Gene for negative selection. Its encodes bacteriotoxin that inhibit growth of E. coli by interfering with E. coli gyrase.
- ◉ **Types of gateway vectors:**
 1. Donor vector. (ccdB+, KanR)
 - i. Denoted by pDONR
 - ii. attP sites
 - iii. Used to clone attB containing PCR-product

GATEWAY CLONING

2. Entry vector: (ccdB-, KanR)

- i. Denoted by pENTR
- ii. attL sites
- iii. Produce when BP reaction take place.

3. Designation vector: (ccdB+, ampR)

- i. denoted by pDEST)
- ii. attR sites
- iii. Provides backbone for expression clone

4. Expression clone: (ampR)

- i. attB sites
- ii. It is final product of gateway cloning.
- iii. Ready to analyse gene of interest. It contain all necessary condition for gene expression.

GATEWAY CLONING

- ⊙ Features of vector.

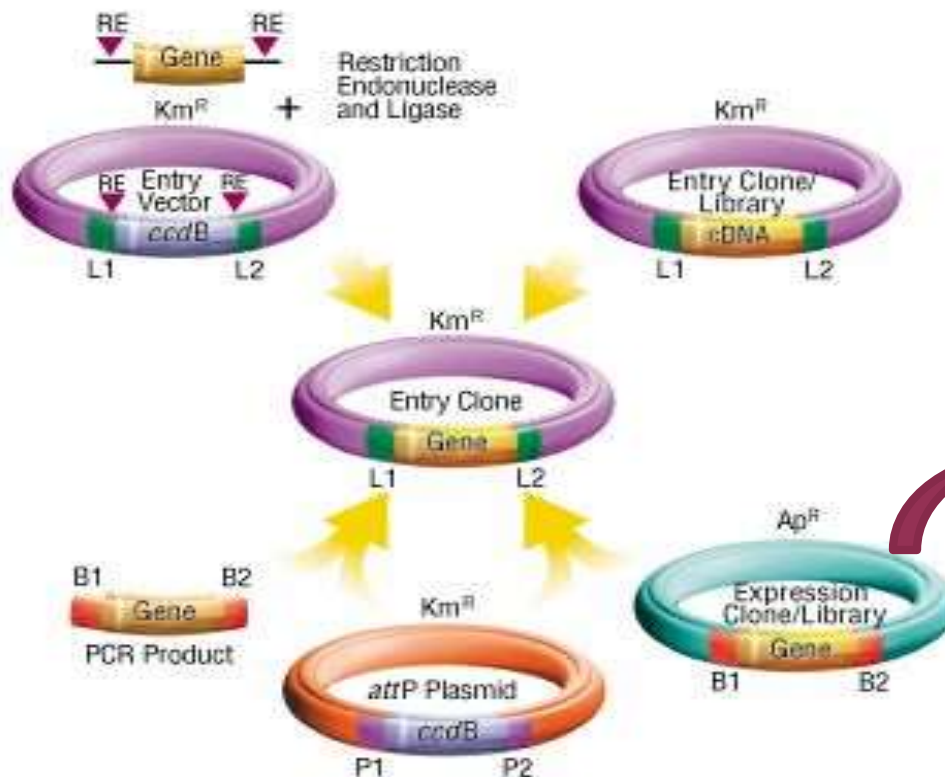
1. Two att sites.
2. ccdB gene
3. Resistance gene

- ⊙ Process of gateway cloning

- a. Insert gene of insert (entry clone form)

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- There are Four ways of obtaining pENTR



An expression vector is usually a plasmid or virus designed for protein expression in cells.

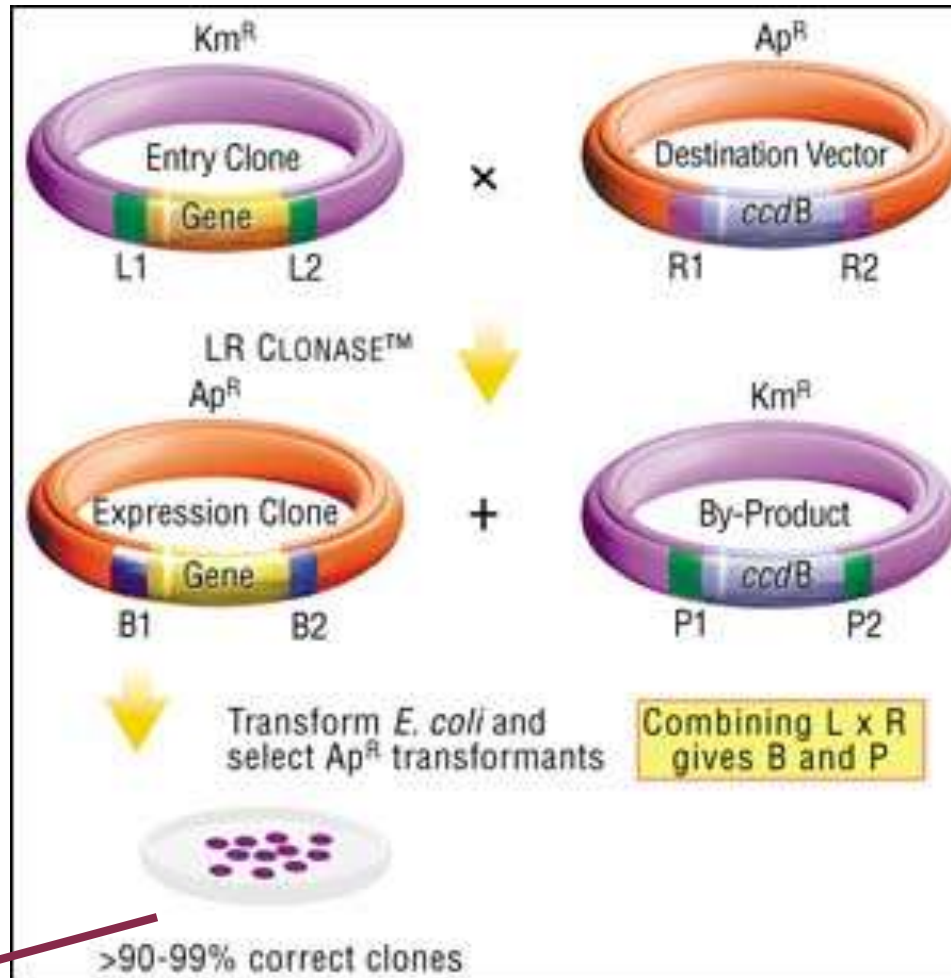
GATEWAY CLONING

- ◉ All entry clone produce have attL sites on both side of gene of interest.
- ◉ The selection of entry clone is done on media , only cell containing kanamycin will grow and also ccdB do negative selection.
- ◉ The attL sites are sites recognized by clonase that cut them and then matchup this entry clone with pDEST which contains attR sites .
- ◉ Process called LR reaction. And expression clone is formed.
- ◉ This expression clone formed actually contain segments of L and R recombination sites, but new site formed is attB in expression clone. And attP in byproduct.

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- ⦿ This process is reversible.
- ⦿ Selection media is used to select expression clone.
- ⦿ Ampicilin is added to media so, only cell containing ampicilin will grow.
- ⦿ The clone obtain in this way are 90-99% correct.

GATEWAY CLONING



Colonies containing expression clone

GATEWAY CLONING

APPLICATIONS:

- ◉ 99% cloning efficiency delivers the clone you need
- ◉ Maintain orientation and reading frame throughout cloning process
- ◉ Efficient cloning of single fragments into multiple vectors simultaneously
- ◉ Flexibility to clone multiple gene fragments into a single construct
- ◉ Allows sub-cloning from one vector backbone to another.
- ◉ Supports site specific recombination.
- ◉ It saves your time.